
CP-MLPs: A Tensor-Rank Theory of Tied and Untied MLP Blocks

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Abstract

Transformer feed-forward blocks combine several design choices: squared activations, gating, and tied versus untied parameter sharing, whose separate contributions to expressivity remain poorly understood. We introduce *CP-MLPs*, a tensor framework in which each multiplicative unit contributes a rank-1 interaction with three roles: detecting an input feature, reading a value, and writing to the output. Tied gated blocks such as ReLU^2 force the detector and value roles to share the same direction; untied gated blocks such as ReGLU and SwiGLU let them differ. We show that this distinction yields exact, degree-wise tensor-rank separations: untied detector-value units are strictly more width-efficient for indefinite quadratic interactions and role-asymmetric monomials, while tied units are structurally matched to symmetric pure-power targets. Tying is therefore not inherently better or worse where its benefit depends on the structure of the target. In experiments, models below the required width cannot represent the target exactly and stall at a loss plateau that does not improve with extra optimization budget. The detector-value advantage also persists in trained gated MLPs at matched parameter count.

1 Introduction

Feed-forward layers account for roughly two-thirds of the parameters in a transformer [Vaswani et al., 2017] language model [Geva et al., 2021], and a wide range of designs have been proposed for them. Common variants include squared activations such as ReLU^2 [So et al., 2021], gated blocks such as ReGLU and SwiGLU [Shazeer, 2020], and smooth square-style variants such as SiLU^2 . Bilinear MLPs are less common as training blocks in modern LLMs, but they provide a clean quadratic base case and have received attention in the interpretability literature [Pearce et al., 2024]. These architectures differ not only in their gating nonlinearity, but also in whether the same direction is used both to detect a feature and to read out its value. In this work we ask when untying these two roles improves expressivity, in the sense of reducing the number of channels required to represent a given interaction.

We introduce *CP-MLPs*, a tensor framework for analyzing feed-forward blocks based on canonical polyadic (CP) tensor decomposition. Each multiplicative unit contributes a rank-1 atom [Kolda and Bader, 2009] with three roles: a *detector* that selects an input feature, a *value* that supplies the signal, and an *output* direction that writes the result. Tied blocks such as ReLU^2 and SiLU^2 use the same direction for the detector and value. Untied blocks such as bilinear MLPs, ReGLU , and SwiGLU allow the two to differ. This decouples two design choices that are typically conflated: the choice of gate, and whether the detector and value branches are tied.

The comparison is naturally made after input symmetrization. A polynomial function evaluates its coefficient tensor on repeated copies of the same input, so only the input-symmetric part of the

tensor contributes to the function it represents [Comon et al., 2008]. Tied and untied blocks therefore parameterize the same space of input-symmetric tensors, but through different families of rank-1 atoms. Untying helps when the target interaction is better matched to an untied detector-value atom than to a tied pure-power atom.

Our analysis gives exact width separations at fixed polynomial degree. At degree two, untied bilinear atoms can represent certain indefinite quadratic interactions with strictly fewer channels than tied atoms. At higher degree, role-asymmetric monomials are realized by a single untied detector-value atom but require multiple tied pure-power atoms. For smooth gates such as SwiGLU, these statements transfer degree by degree to the Taylor coefficient tensors. The point is not that tying is uniformly worse: tied atoms are well suited to symmetric pure-power targets, and untied atoms to interactions that separate detection from value.

Our experiments confirm that these rank separations translate into observable loss differences. In coefficient space, widths below the predicted atom rank produce a persistent loss plateau: the target lies outside the model class at that width, and additional optimization does not close the gap. The low-width advantage is target-dependent—tied units fit pure-power targets directly, while untied detector-value units fit role-asymmetric targets directly—and the untied advantage persists in trained gated MLPs, including under a matched-parameter control.

Contributions.

1. **A unified CP-MLP framework.** We express bilinear, ReLU^2 , SiLU^2 , ReLU , and SwiGLU blocks in a shared atom-based tensor notation with detector, value, and output roles.
2. **Exact atom-rank separations.** We prove that untied detector-value atoms achieve strictly lower width on indefinite quadratic interactions and role-asymmetric monomials, while tied atoms remain well matched to pure-power targets.
3. **Empirical link between rank and loss.** We verify the predicted width thresholds in coefficient space, observe a persistent loss plateau below them, and confirm that the untied advantage carries over to trained gated MLPs at matched parameter count.

2 Related Works

Transformer MLP variants and interpretations. Transformer feed-forward designs have diversified considerably. GLU-style activations were introduced for language modeling by [Dauphin et al., 2017], and variants such as ReLU and SwiGLU were later shown to improve Transformer feed-forward layers [Shazeer, 2020]. Squared- ReLU activations emerged as effective Transformer primitives through architecture-search studies [So et al., 2021], building on a longer line of activation-function design that includes Swish/SiLU [Ramachandran et al., 2017]. In parallel, interpretability work has argued that Transformer feed-forward layers behave like structured key-value memories rather than generic nonlinear post-processing [Geva et al., 2021]. These works motivate studying MLP blocks as structured computational modules, but neither isolates the expressivity effect of tying versus untying detector and value branches, which is the focus of this paper.

Tensor, polynomial, and Waring-rank views. Our framework uses canonical polyadic (CP) tensor decomposition, where tensors are written as sums of rank-1 outer products [Kolda and Bader, 2009]. Recent work on bilinear MLPs also exploits tensor structure to analyze feed-forward blocks, observing that bilinear layers admit a third-order tensor representation [Pearce et al., 2024]. The Π -Net family of [Chrysos et al., 2020, 2021] parameterizes polynomial expansions of the input directly through tensor decompositions, demonstrating that such polynomial blocks are competitive with standard activations on a range of tasks. Polynomial networks have also been studied with algebraic and tensor tools, for example through the geometry of polynomial function classes [Kileel et al., 2019]. Our work differs in that we do not propose polynomial networks as an architecture; we use polynomial structure as an analytical lens for comparing existing MLP variants.

For the higher-degree separation we rely on classical Waring rank: a scalar tied degree- k channel is a pure power $(\ell^\top x)^k$, so tied atom rank coincides with Waring rank. The monomial Waring-rank formula of [Carlini et al., 2011], together with the real-rank result of [Carlini et al., 2017], implies that $x_1^{k-1}x_2$ requires k tied atoms but only one untied detector-value atom. Our contribution is to

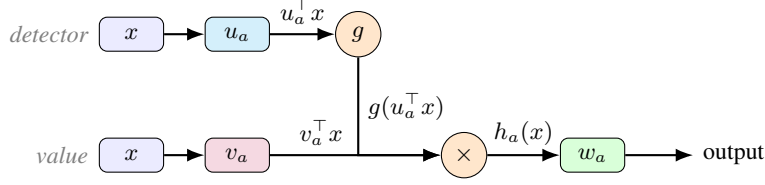


Figure 1: **Leg diagram of a single CP-MLP channel.** The channel computes $h_a(x) = g(u_a^\top x)(v_a^\top x)$ and writes the result through w_a . At each active polynomial degree, the channel contributes one rank-1 CP atom with output, detector, and value legs.

connect these rank facts to practical MLP atom families, showing that tied and untied blocks draw on different atoms inside the same input-symmetric tensor space.

3 The CP-MLP framework

We use a common tensor notation for the multiplicative feed-forward blocks studied in this paper. Bilinear blocks multiply two linear projections, GLU-style blocks gate one projection by another, and square-style blocks reuse the same projection twice. In every case, each hidden channel produces a product of projected input directions, and the output matrix sums these channel-wise products. This is exactly the structure of a canonical polyadic (CP) decomposition [Kolda and Bader, 2009], and we call the resulting representation a *CP-MLP*.

The framework separates two design choices: *the gate*, which selects the polynomial degrees that appear in a local expansion, and *the tying pattern*, which selects whether the detector and value directions of a channel are shared. The expressivity analysis in Section 4 isolates the effect of the second choice.

Definition 3.1 (CP-MLP block). *Fix a scalar gate $g : \mathbb{R} \rightarrow \mathbb{R}$ and let $G : \mathbb{R}^m \rightarrow \mathbb{R}^m$ be its coordinatewise extension, $G(z) = (g(z_1), \dots, g(z_m))^\top$. Given weights $U, V \in \mathbb{R}^{m \times d}$ (rows $u_a^\top, v_a^\top$) and $W \in \mathbb{R}^{p \times m}$ (columns w_a), a width- m CP-MLP block is*

$$F(x) = W(G(Ux) \odot Vx) = \sum_{a=1}^m w_a g(u_a^\top x)(v_a^\top x).$$

We call u_a the detector direction, v_a the value direction, and w_a the output direction of channel a . The block is tied if $V = U$, and untied otherwise.

The detector direction selects the scalar feature passed through the gate, while the value direction supplies the linear signal that the gate modulates. Tying forces both roles onto the same direction; untying lets them differ. Figure 1 shows the corresponding three-leg view of a single channel.

3.1 Degree-wise atoms

Assume the gate is analytic at the origin, $g(z) = \sum_{j \geq 0} c_j z^j$ (piecewise-analytic gates such as ReLU are treated region-wise; see Table 1). Substituting $z = u_a^\top x$ and multiplying by $v_a^\top x$, one channel contributes $\sum_{j \geq 0} c_j (u_a^\top x)^j (v_a^\top x)$; reindexing by total degree $k = j + 1$, the degree- k term carries the coefficient c_{k-1} . Writing $T[x, \dots, x] \in \mathbb{R}^p$ for the contraction of the k input modes of $T \in \mathbb{R}^p \otimes (\mathbb{R}^d)^{\otimes k}$ against x (output mode free; conventions in Appendix A.1), the degree- k component of the block is $F^{(k)}(x) = T^{(k)}[x, \dots, x]$, with raw coefficient tensor

$$T^{(k)} = c_{k-1} \sum_{a=1}^m w_a \otimes u_a^{\otimes (k-1)} \otimes v_a \in \mathbb{R}^p \otimes (\mathbb{R}^d)^{\otimes k}.$$

Each active degree thus admits a CP decomposition with at most m rank-one terms, one per channel. We define $\mathcal{S}(g)$ for the *polynomial signature* of the gate, i.e. the set of active degrees.

Definition 3.2 (Polynomial signature). *The polynomial signature of g is $\mathcal{S}(g) := \{k \geq 1 : c_{k-1} \neq 0\}$.*

Table 1: Feed-forward blocks as CP-MLPs. The gate controls which degrees are active through its polynomial signature; the tying pattern controls whether detector and value directions are shared.

Block	Gate $g(z)$	Tying	Signature $\mathcal{S}(g)$	Effective atom
Bilinear [Pearce et al., 2024]	z	untied	$\{2\}$	$w \otimes \text{Sym}(u \otimes v)$
ReGLU [Shazeer, 2020]	$\text{ReLU}(z)$	untied	$\{2\}^\dagger$	region-wise $w \otimes \text{Sym}(u \otimes v)$
SwiGLU [Shazeer, 2020]	$\text{SiLU}(z) = z\sigma(z)$	untied	$\{2, 3, 5, 7, \dots\}$	$w \otimes \text{Sym}(u^{\otimes(k-1)} \otimes v)$
ReLU^2 [So et al., 2021]	$\text{ReLU}(z)$	tied	$\{2\}^\dagger$	region-wise $w \otimes u^{\otimes 2}$
SiLU^2	$z\sigma(z)^2$	tied	$\{2, 3, 4, 5, \dots\}$	$w \otimes u^{\otimes k}$

[†]Region-wise: ReLU is piecewise linear, so the channel is a degree-2 polynomial inside each fixed activation region.

The gate selects which degrees are active through $\mathcal{S}(g)$; the tying pattern selects the atom used at each active degree. When $v_a = u_a$, the raw degree- k atom collapses to $w_a \otimes u_a^{\otimes k}$; when v_a is independent of u_a , it retains the detector-value form $w_a \otimes u_a^{\otimes(k-1)} \otimes v_a$.

3.2 Effective atoms after symmetrization

The tensor above distinguishes detector and value slots, but the function it represents evaluates the tensor on repeated copies of the same input, so input-slot order is not observable. We write $\text{Sym}(T)$ for the symmetrization of T , the average of T over all permutations of its input modes (with the output mode, when present, left fixed), then

$$T[x, \dots, x] = \text{Sym}(T)[x, \dots, x].$$

The proof is immediate: every permutation of T receives the same vector x in every input slot. A full statement, together with the quadratic matrix specialization, is given in Appendix B.1.

After symmetrization, what remains is the set of symmetric rank-1 tensors that each architecture can build—one set for tied blocks and one for untied.

Definition 3.3 (Effective tied and untied CP-MLP atoms). *The degree- k effective untied atom is*

$$\mathcal{A}_{\text{untied}}^{(k)}(w, u, v) := w \otimes \text{Sym}(u^{\otimes(k-1)} \otimes v) \in \mathbb{R}^p \otimes \text{Sym}^k(\mathbb{R}^d),$$

where $\text{Sym}^k(\mathbb{R}^d)$ denotes the space of symmetric order- k tensors (Definition B.1). The degree- k effective tied atom is the specialization $v = u$:

$$\mathcal{A}_{\text{tied}}^{(k)}(w, u) := w \otimes u^{\otimes k} \in \mathbb{R}^p \otimes \text{Sym}^k(\mathbb{R}^d).$$

Tied and untied blocks therefore parameterize tensors in the same symmetric space $\mathbb{R}^p \otimes \text{Sym}^k(\mathbb{R}^d)$, but draw on different atom families: tied blocks use pure-power atoms $w \otimes u^{\otimes k}$, while untied blocks use symmetrized detector-value atoms $w \otimes \text{Sym}(u^{\otimes(k-1)} \otimes v)$. Setting $v = u$ recovers the tied family as a special case. The benefit of untying is not that antisymmetric components become visible, but that the symmetrized untied family can represent certain symmetric tensors with strictly fewer atoms.

Table 1 places the blocks studied in this paper along the two axes of the framework: the gate and the tying pattern. Bilinear, ReGLU, and SwiGLU differ only in their gates and share the untied detector-value atom structure. ReLU^2 and SiLU^2 instead use tied atoms, since the same direction supplies both the detector and value roles. The next section asks how many atoms of each family are required to represent a given target tensor.

4 Expressivity via CP-MLP atom rank

The atom families of Section 3.2 suggest a natural notion of width: the smallest number of atoms needed to express a given coefficient tensor. Fixing an active degree k , each CP-MLP channel contributes one effective atom in $\mathcal{T}_k := \mathbb{R}^p \otimes \text{Sym}^k(\mathbb{R}^d)$, so the question we want to answer is how many such atoms are required to represent a target $T \in \mathcal{T}_k$. We call this minimum the *CP-MLP atom*

rank; since each atom corresponds to one channel at degree k , it is the degree-wise width required to represent a coefficient tensor.

For polynomial gates, $T^{(k)}$ is the ordinary degree- k coefficient tensor; for analytic gates, it is the degree- k Taylor coefficient tensor, satisfying $D^k F(0) = k! \text{Sym}(T^{(k)})$. Throughout this section k denotes an active degree of the gate, and any nonzero scalar Taylor coefficient is absorbed into the output factor w , so the rank comparisons below depend only on the channel atom family.

4.1 Atom-rank definitions

Definition 4.1 (Tied CP-MLP atom rank). *For $T \in \mathcal{T}_k$, the tied CP-MLP atom rank $r_{\text{tied}}(T)$ is the smallest integer m such that $T = \sum_{a=1}^m w_a \otimes u_a^{\otimes k}$, $w_a \in \mathbb{R}^p$, $u_a \in \mathbb{R}^d$.*

Definition 4.2 (Untied CP-MLP atom rank). *For $T \in \mathcal{T}_k$, the untied CP-MLP atom rank $r_{\text{untied}}(T)$ is the smallest integer m such that $T = \sum_{a=1}^m w_a \otimes \text{Sym}(u_a^{\otimes(k-1)} \otimes v_a)$, $w_a \in \mathbb{R}^p$, $u_a, v_a \in \mathbb{R}^d$.*

Both ranks are minimum-width quantities at a fixed degree, counting the fewest atoms required to express a coefficient tensor in the corresponding family. Since a tied atom is recovered from an untied atom by setting $v_a = u_a$, the untied family contains the tied family, which immediately implies the following inequality.

Proposition 4.3 (Untied atoms contain tied atoms). *For every $T \in \mathcal{T}_k$, $r_{\text{untied}}(T) \leq r_{\text{tied}}(T)$.*

Proof. Any tied representation $T = \sum_a w_a \otimes u_a^{\otimes k}$ is also an untied representation with $v_a = u_a$, since $u_a^{\otimes k} = \text{Sym}(u_a^{\otimes(k-1)} \otimes u_a)$. Minimizing over tied representations yields the claim. $\square$

The remainder of the section asks when this inequality is strict. We first study degree two, where the coefficient tensor is a symmetric matrix, and then prove a higher-degree separation using a role-asymmetric monomial.

Remark 4.4 (Scalar targets suffice). *For a scalar target $S \in \text{Sym}^k(\mathbb{R}^d)$, embedding $T = e_o \otimes S$ gives $r_{\text{tied}}(T) = r_{\text{tied}}(S)$ and $r_{\text{untied}}(T) = r_{\text{untied}}(S)$: output projection cannot increase atom rank, and any scalar representation lifts by tensoring with e_o . Full details are in Appendix C.1. The separations below are stated for scalar targets.*

4.2 Degree-two bilinear atoms

We first specialize to degree 2, where a scalar coefficient tensor is a symmetric matrix and the atom-rank question reduces to representing quadratic forms

$$f_M(x) = x^\top M x, \quad M = M^\top \in \mathbb{R}^{d \times d}.$$

A tied atom contributes the signed square $\alpha(u^\top x)^2$, corresponding to the rank-1 symmetric matrix $\alpha u u^\top$. An untied atom contributes a product of two linear forms, $\alpha(u^\top x)(v^\top x)$, which after symmetrization becomes

$$\frac{\alpha}{2}(u v^\top + v u^\top).$$

A tied atom therefore contributes one quadratic direction, while an untied atom can couple two. Our first result records this coarse rank comparison.

Theorem 4.5 (Coarse bilinear rank bounds). *Let $M \in \mathbb{R}^{d \times d}$ be symmetric. If M is represented by a tied bilinear block of width m , then $\text{rank}(M) \leq \min(m, d)$. If M is represented by an untied bilinear block of width m , then $\text{rank}(M) \leq \min(2m, d)$. Both bounds are tight.*

Proof sketch. Each tied unit contributes the rank-1 symmetric matrix $u_a u_a^\top$, so a width- m tied block produces matrices of rank at most $\min(m, d)$. For untied blocks, only the symmetric part of $u_a v_a^\top$ contributes; the resulting matrix $\frac{1}{2}(u_a v_a^\top + v_a u_a^\top)$ has image in $\text{span}\{u_a, v_a\}$ and therefore rank at most two, and summing m such terms gives the bound. Both bounds are tight by choosing independent coordinate directions. $\square$

Theorem 4.5 shows that an untied atom can carry twice the rank of a tied atom, but rank alone does not determine which matrices a given width can represent. The rank-2 gain has a specific sign structure. From the difference-of-squares identity

$$(u^\top x)(v^\top x) = \frac{1}{4}((u+v)^\top x)^2 - \frac{1}{4}((u-v)^\top x)^2,$$

a non-collinear untied atom is a difference of two squares. It can pair one positive with one negative quadratic direction—representing interactions like $a^2 - b^2 = (a+b)(a-b)$ —but cannot compress two same-sign directions such as $a^2 + b^2$ into a single atom. The exact width is therefore controlled not by rank, but by how many positive and negative quadratic directions the target carries.

Let $n_+(M)$ and $n_-(M)$ denote the numbers of positive and negative eigenvalues of M .

Theorem 4.6 (Exact bilinear atom ranks). *For every symmetric matrix $M \in \mathbb{R}^{d \times d}$,*

$$r_{\text{tied}}(M) = \text{rank}(M) = n_+(M) + n_-(M),$$

whereas

$$r_{\text{untied}}(M) = \max\{n_+(M), n_-(M)\}.$$

Proof sketch. For tied atoms, each channel contributes one signed rank-1 square, so it needs one atom per nonzero eigendirection, and the spectral decomposition achieves this number. For untied atoms, the difference-of-squares identity expresses any width- m representation as $M = AA^\top - BB^\top$ with $A, B \in \mathbb{R}^{d \times m}$. On $\ker(A^\top)$, $x^\top Mx = -\|B^\top x\|^2 \leq 0$, so M cannot be positive definite on a subspace of dimension greater than m ; the analogous argument on $\ker(B^\top)$ gives the same bound for negative directions, so $m \geq \max\{n_+(M), n_-(M)\}$. The matching upper bound is achieved by pairing opposite-sign eigendirections one atom at a time, with unpaired directions handled by the tied special cases $u = \pm v$. Full proofs of Theorems 4.5 and 4.6 are in Appendix C.2. $\square$

Corollary 4.7 (When untied bilinear atoms help). *If $M \succeq 0$ or $M \preceq 0$,*

$$r_{\text{tied}}(M) = r_{\text{untied}}(M) = \text{rank}(M).$$

If M is indefinite,

$$r_{\text{tied}}(M) - r_{\text{untied}}(M) = \min\{n_+(M), n_-(M)\}.$$

Untied atoms are therefore most width-efficient on indefinite quadratic interactions; they offer no advantage on purely same-sign quadratic structure.

Connection to hard-gated blocks. The same degree-two atom gap appears regionwise in hard-gated CP-MLPs. On a fixed activation pattern $\mu \in \{0, 1\}^m$, ReGLU realizes $Q_{\mu, o} = \sum_{a=1}^m \mu_a W_{oa} \frac{u_a v_a^\top + v_a u_a^\top}{2}$ for each output coordinate o , whereas ReLU² realizes $Q_{\mu, o} = \sum_a \mu_a W_{oa} u_a u_a^\top$. Inside each fixed region, ReGLU has access to untied rank-2 atoms with opposite-sign structure, while ReLU² is restricted to tied rank-1 atoms. Region switching adds further piecewise expressivity; Theorem 4.6 isolates the within-region atom difference.

4.3 Higher-degree monomial separation

The degree-two analysis showed that untied atoms can be more width-efficient when the target contains opposite-sign quadratic structure. At higher degree, a different kind of separation emerges that is driven not by sign, but by *role asymmetry*: one direction is repeated many times in the detector slot, while another direction appears only once in the value slot.

For $k \geq 2$, consider the binary monomial $F_k(x_1, x_2) = x_1^{k-1}x_2$. This target has the exact shape of a single untied atom: taking $u = e_1$ and $v = e_2$ gives $(u^\top x)^{k-1}(v^\top x) = x_1^{k-1}x_2$, so one untied atom suffices. A tied degree- k atom, by contrast, has the pure-power form $(\ell^\top x)^k$, so the tied atom rank of F_k is exactly its Waring rank.

Definition 4.8 (Waring rank). *Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and let $f : \mathbb{F}^d \rightarrow \mathbb{F}$ be a homogeneous polynomial of degree k . The Waring rank of f over $\mathbb{F}$, denoted $\text{Wr}_{\mathbb{F}}(f)$, is the smallest integer m such that*

$$f(x) = \sum_{a=1}^m c_a \ell_a(x)^k,$$

where $c_a \in \mathbb{F}$ and each ℓ_a is a linear form over $\mathbb{F}$.

For scalar tied CP-MLPs, the real Waring rank coincides with the tied atom rank: any tied representation $f(x) = \sum_a c_a (u_a^\top x)^k$ with $u_a \in \mathbb{R}^d$ is exactly a real Waring decomposition, so $r_{\text{tied}}(f) = \text{Wr}_{\mathbb{R}}(f)$ for every scalar homogeneous degree- k polynomial. This is the standard correspondence between symmetric tensor rank and polynomial Waring rank [Comon et al., 2008]. The following classical formula computes the right-hand side for monomials.

Proposition 4.9 (Waring rank of monomials; [Carlini et al., 2011, 2017]). *Let $\mu(x) = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$ be a monomial of degree $k = \sum_i \alpha_i$, with sorted exponents $1 \leq \alpha_1 \leq \cdots \leq \alpha_n$. Then*

$$\text{Wr}_{\mathbb{C}}(\mu) = \frac{1}{\alpha_1 + 1} \prod_{i=1}^n (\alpha_i + 1) = \prod_{i=2}^n (\alpha_i + 1),$$

and $\text{Wr}_{\mathbb{R}}(\mu) = \text{Wr}_{\mathbb{C}}(\mu)$ whenever $\alpha_1 = 1$.

The complex formula is [Carlini et al., 2011, Prop. 3.1]; the real-complex equality for $\alpha_1 = 1$ follows from [Carlini et al., 2017]. For $F_k = x_1^{k-1} x_2$ the sorted exponent vector is $(1, k-1)$, so

$$\text{Wr}_{\mathbb{R}}(F_k) = \text{Wr}_{\mathbb{C}}(F_k) = (k-1) + 1 = k.$$

Theorem 4.10 (Degree- k monomial separation). *For every $k \geq 2$,*

$$r_{\text{untied}}(F_k) = 1, \quad r_{\text{tied}}(F_k) = k.$$

A single untied atom therefore represents a degree- k coefficient tensor that requires k tied atoms.

Proof. We bound the two ranks separately.

Untied rank. Setting $u = e_1$ and $v = e_2$, $(u^\top x)^{k-1} (v^\top x) = x_1^{k-1} x_2 = F_k(x)$, so a single untied atom realizes F_k and $r_{\text{untied}}(F_k) \leq 1$. Since $F_k \neq 0$, it cannot be represented with zero atoms, giving $r_{\text{untied}}(F_k) = 1$.

Tied rank. Any scalar tied degree- k representation $F_k(x) = \sum_a c_a \ell_a(x)^k$ is a real Waring decomposition, so $r_{\text{tied}}(F_k) = \text{Wr}_{\mathbb{R}}(F_k) = k$ by Proposition 4.9 $\square$

Interpretation. The separation does not say that untied atoms are uniformly better. It says they are well matched to interactions with distinct detector and value roles. The monomial $x_1^{k-1} x_2$ is a single atom in the untied family with $u = e_1$ and $v = e_2$; in the tied family of pure powers $(\ell^\top x)^k$, the same target has Waring rank k . The classical Waring-rank obstruction therefore translates into an architectural width separation between tied and untied CP-MLP atoms.

Connection to smooth gates. Theorem 4.10 is stated for homogeneous degree- k atoms, but the degree-wise CP expansion of Section 3.1 carries it over to smooth gates: an untied gated channel contributes $c_{k-1} (u^\top x)^{k-1} (v^\top x)$ to its degree- k Taylor coefficient whenever $c_{k-1} \neq 0$, so the tied-versus-untied atom-rank comparison applies at each degree $k \in \mathcal{S}(g)$ of the gate’s polynomial signature (Table 1). The statement is degree-wise, not a global rank claim for the full analytic function—all active degrees share the same directions, and the gate fixes the Taylor coefficients.

5 Experiments

We test whether the atom-rank separations from Section 4 appear as observable width and loss effects. Indefinite quadratics realize the degree-two gap (Section 5.1); role-asymmetric monomials realize the higher-degree gap (Section 5.2); pure-power targets show that tying can be the better match (Section 5.3); and gated MLPs show that the untied advantage persists at matched parameter count (Section 5.4). Optimization, hyperparameters, and budget checks are in Appendix D.

5.1 Indefinite quadratic targets

Theorem 4.6 predicts that untied bilinear atoms are more width-efficient than tied atoms only when the target carries both positive and negative directions. To test this, we construct

$$Q = \text{diag}(\mathbf{1}^{n_+}, -\mathbf{1}^{n_-}, \mathbf{0}^{d-n_+-n_-}), \quad y(x) = x^\top Q x,$$

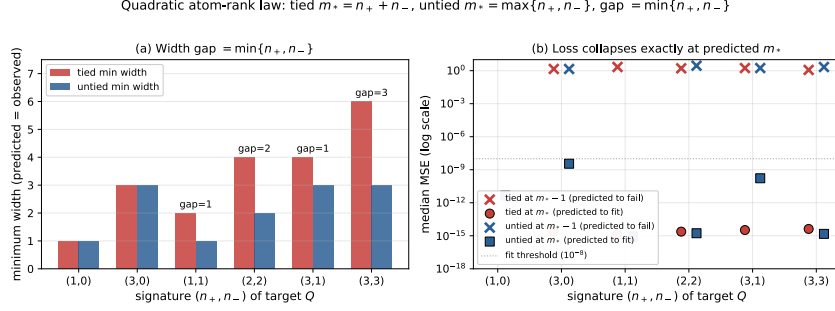


Figure 2: Quadratic targets with (n_+, n_-) positive and negative directions. *Left*: required widths match the predictions $n_+ + n_-$ (tied) and $\max\{n_+, n_-\}$ (untied). *Right*: loss stays near the target variance below the predicted width and reaches machine precision at the predicted width.

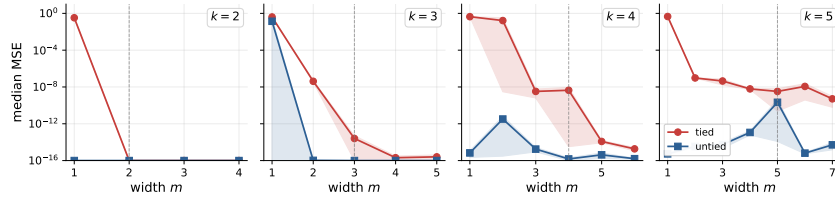


Figure 3: Coefficient-space width sweep for $F_k = x_1^{k-1}x_2$, $k \in \{2, 3, 4, 5\}$. Untied atoms fit at width $m = 1$. Tied atoms fit only at the predicted width $m = k$. The nonzero loss below $m = k$ is the plateau at widths below the atom rank.

with $d = 8$, and sweep six choices of (n_+, n_-) covering definite, balanced indefinite, and imbalanced indefinite targets.

Figure 2 confirms the exact threshold: tied blocks require width $n_+ + n_-$, untied blocks require width $\max\{n_+, n_-\}$. At one width below the prediction, the loss stays near the target variance; at the predicted width, the loss drops to machine precision. Definite targets show no tied/untied gap, while indefinite targets show the predicted gap $\min\{n_+, n_-\}$.

5.2 Role-asymmetric monomial targets

We next test the higher-degree separation of Theorem 4.10. For

$$F_k(x_1, x_2) = x_1^{k-1}x_2, \quad k \in \{2, 3, 4, 5\},$$

the theory predicts untied atom rank 1 and tied atom rank k . We fit the target directly in monomial coefficient space and sweep width $m = 1, \dots, k + 2$; coefficient-space fitting removes sampling noise and isolates the rank effect.

Figure 3 shows sharp thresholds at the predicted ranks. Untied atoms reach machine precision at $m = 1$ for every degree, while tied atoms remain at a nonzero plateau for $m < k$ and collapse at $m = k$. Below its atom rank the target lies outside the tied model class, but for $2 \leq m < k$ it lies in the class's closure: the border Waring rank of $x_1^{k-1}x_2$ is two [Comas and Seiguer, 2011], via $x_1^{k-1}x_2 = \lim_{\epsilon \rightarrow 0} \frac{1}{k\epsilon} [(x_1 + \epsilon x_2)^k - x_1^k]$. The plateau therefore reflects the ill-conditioning of border-rank approximations, whose coefficients must diverge as the residual shrinks, rather than a positive distance to the model class; at $m = 1$ the floor is genuine, since the set of symmetric rank-at-most-one tensors is closed. The untied advantage is specific to single-value-role targets; products with multiple independent value roles require more atoms (Appendix D.7).

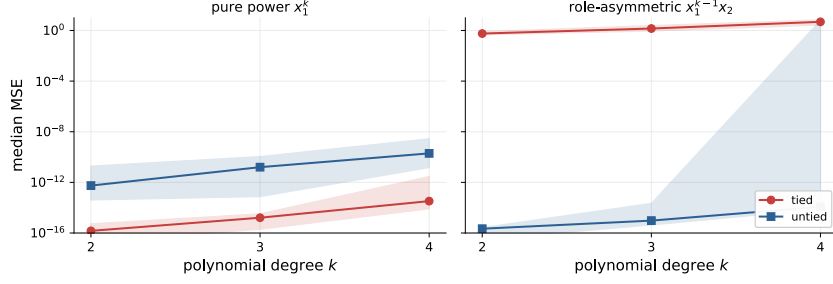


Figure 4: Target structure changes the low-width winner. At $m = 1$, tied atoms fit pure powers x_1^k , while untied atoms fit role-asymmetric monomials $x_1^{k-1}x_2$. Bands show min and max over seeds.

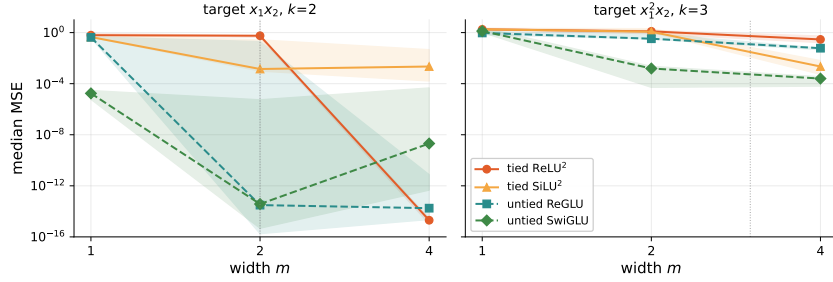


Figure 5: Width versus loss for actual gates on $x_1^{k-1}x_2$. *Left*: for $k = 2$, untied ReGLU and SwiGLU reach near-machine precision at smaller width than tied ReLU^2 and SiLU^2 . *Right*: for $k = 3$, the exact rank threshold becomes a qualitative ordering; SwiGLU is best and untied models remain below tied models across tested widths.

5.3 Pure powers versus role-asymmetric targets

The previous experiment shows a strong untied advantage on role-asymmetric monomials, but tying is not always worse. At width $m = 1$, we compare tied and untied models on two targets:

$$x_1^k \quad \text{and} \quad x_1^{k-1}x_2.$$

The first is a pure power and is realized exactly by one tied atom; the second is role-asymmetric and is realized exactly by one untied atom. Figure 4 shows the expected reversal: on pure powers, tied models reach machine precision while untied models must align the redundant value direction with the detector; on role-asymmetric targets, untied models fit at width one while tied models cannot. The low-width winner therefore depends on target structure.

5.4 Separation in gated MLPs

The coefficient-space experiments above isolate the atom-rank effect; we now test actual gated MLPs on the same witness $F_k = x_1^{k-1}x_2$. We compare tied ReLU^2 , tied SiLU^2 , untied ReGLU, and untied SwiGLU, using Gaussian inputs with $d = 4$, $k \in \{2, 3\}$, and widths $m \in \{1, 2, 4\}$.

Figure 5 shows that the exact polynomial threshold becomes a stable qualitative ordering for gated networks. At $k = 2$, untied ReGLU and SwiGLU reach near-machine precision at $m = 2$, while tied ReLU^2 requires $m = 4$ and tied SiLU^2 plateaus. At $k = 3$, no architecture reaches machine precision because the relevant term comes from higher Taylor coefficients, but SwiGLU remains the best-performing family and untied models lie below tied models at every tested width.

A matched-parameter control comparing untied width m against tied width $2m$ preserves the untied advantage at both degrees, with untied models reaching one to several orders of magnitude lower loss than their tied parameter-matched counterparts; the matched-parameter figure and full numerical table are in Appendix D.8.

6 Conclusion

We introduced CP-MLPs, a tensor framework for comparing tied and untied multiplicative feed-forward blocks. The framework separates the gate from the choice of whether detector and value directions are shared. Our theory shows that this choice changes width efficiency. Untied atoms can represent indefinite quadratic interactions and role-asymmetric monomials with fewer channels, while tied atoms are naturally matched to pure-power targets. For smooth gates such as SwiGLU, these statements apply degree by degree to Taylor coefficient tensors. Experiments confirm the predicted width thresholds, show loss plateaus below the required rank, and demonstrate that the low-width advantage changes with target structure. The same untied detector-value advantage remains visible in trained gated MLPs at matched parameter count. These results suggest that the success of GLU-style feed-forward blocks is not only an activation effect: separating detector and value directions changes which interactions are width-efficient. Several open directions remain. Our higher-degree separation is degree-wise rather than a global statement for the full analytic channel, and our experiments use synthetic polynomial targets rather than language-modeling tasks; bridging both gaps, by extending the theory to richer multi-leg atoms and by testing these effects in full Transformer language models, is a natural next step.

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A Preliminaries

This appendix collects the tensor and CP-decomposition notation used throughout the paper.

A.1 Tensors and CP decompositions

Let $T \in \mathbb{R}^{d_1 \times \dots \times d_n}$ be an order- n tensor and let $x_i \in \mathbb{R}^{d_i}$. We write the multilinear contraction of T with vectors $x_1, \dots, x_n$ as

$$T[x_1, \dots, x_n] := T \times_1 x_1 \times_2 \dots \times_n x_n,$$

where $\times_i$ denotes contraction along mode i :

$$(T \times_i x_i)_{j_1, \dots, j_{i-1}, j_{i+1}, \dots, j_n} = \sum_{k=1}^{d_i} T_{j_1, \dots, j_{i-1}, k, j_{i+1}, \dots, j_n} (x_i)_k.$$

Equivalently,

$$T[x_1, \dots, x_n] = \sum_{j_1=1}^{d_1} \dots \sum_{j_n=1}^{d_n} T_{j_1, \dots, j_n} (x_1)_{j_1} \dots (x_n)_{j_n}.$$

When all modes are contracted with vectors, the result is a scalar. If some modes are left uncontracted, the same formula applies to the remaining modes only; in particular, contracting the k input modes of $T \in \mathbb{R}^p \otimes (\mathbb{R}^d)^{\otimes k}$ while leaving the output mode free gives $T[x_1, \dots, x_k] \in \mathbb{R}^p$, the convention used in Section 3.1. We write e_i for the i -th standard basis vector.

A canonical polyadic (CP) decomposition represents a tensor as a sum of rank-one outer products:

$$T = \sum_{a=1}^r \lambda_a u_a^{(1)} \otimes u_a^{(2)} \otimes \dots \otimes u_a^{(n)},$$

where $u_a^{(i)} \in \mathbb{R}^{d_i}$ and $\lambda_a \in \mathbb{R}$. Under contraction, each rank-one term factorizes:

$$T[x_1, \dots, x_n] = \sum_{a=1}^r \lambda_a \prod_{i=1}^n \langle u_a^{(i)}, x_i \rangle.$$

Thus a CP decomposition expresses a multilinear map as a sum of separable products of linear forms.

This is the tensor structure used throughout the paper. For example, a single CP-MLP channel at degree k contributes a rank-one raw tensor of the form

$$w_a \otimes u_a^{\otimes(k-1)} \otimes v_a,$$

whose diagonal evaluation gives

$$(w_a \otimes u_a^{\otimes(k-1)} \otimes v_a)[x, \dots, x] = w_a (u_a^\top x)^{k-1} (v_a^\top x).$$

A width- m CP-MLP sums m such channel-wise rank-one contributions.

Symmetrization. For any tensor $T \in (\mathbb{R}^d)^{\otimes k}$, define

$$\text{Sym}(T) = \frac{1}{k!} \sum_{\pi \in S_k} \pi \cdot T.$$

Symmetric CP rank. A symmetric tensor T has symmetric rank r if r is the smallest integer such that

$$T = \sum_{a=1}^r \lambda_a u_a^{\otimes k}.$$

Matricization lower bound. For any tensor T , the rank of any flattening provides a lower bound on its CP rank [Kolda and Bader, 2009].

B Additional framework details

B.1 Symmetrization details

Let S_k denote the symmetric group on k elements. For a permutation $\pi \in S_k$ and a tensor $T \in \mathbb{R}^p \otimes (\mathbb{R}^d)^{\otimes k}$, write $\pi \cdot T$ for the tensor obtained by permuting the k input modes according to π , leaving the output mode fixed.

Definition B.1 (Input-symmetric tensor). *A tensor*

$$T \in \mathbb{R}^p \otimes (\mathbb{R}^d)^{\otimes k}$$

is input-symmetric if $\pi \cdot T = T$ for every $\pi \in S_k$. The space of input-symmetric tensors is denoted

$$\mathbb{R}^p \otimes \text{Sym}^k(\mathbb{R}^d).$$

Definition B.2 (Input symmetrization). *The input symmetrization of $T \in \mathbb{R}^p \otimes (\mathbb{R}^d)^{\otimes k}$ is*

$$\text{Sym}(T) := \frac{1}{k!} \sum_{\pi \in S_k} \pi \cdot T.$$

Lemma B.1 (Diagonal evaluation sees only the input-symmetric part). *For every*

$$T \in \mathbb{R}^p \otimes (\mathbb{R}^d)^{\otimes k}$$

and every $x \in \mathbb{R}^d$,

$$T[x, \dots, x] = \text{Sym}(T)[x, \dots, x].$$

Proof. For any $\pi \in S_k$, all input slots of $\pi \cdot T$ receive the same vector x . Hence

$$(\pi \cdot T)[x, \dots, x] = T[x, \dots, x].$$

Averaging over all $\pi \in S_k$ gives the claim. □

Lemma B.2 (Quadratic diagonal evaluation). *For every $A \in \mathbb{R}^{d \times d}$ and every $x \in \mathbb{R}^d$,*

$$x^\top A x = x^\top \frac{A + A^\top}{2} x.$$

Proof. Let

$$S = \frac{1}{2}(A - A^\top).$$

Then $S^\top = -S$. Since $x^\top S x$ is a scalar,

$$x^\top S x = (x^\top S x)^\top = x^\top S^\top x = -x^\top S x.$$

Therefore $x^\top S x = 0$. The result follows by writing

$$A = \frac{1}{2}(A + A^\top) + S.$$

□

Derivative tensors. For an analytic scalar function, the degree- k Taylor coefficient tensor is equivalently the k -th derivative tensor up to the factor $k!$. In particular, for a homogeneous tensor polynomial

$$f_T(x) = T[x, \dots, x],$$

we have

$$D^k f_T(0) = k! \text{Sym}(T).$$

Thus the derivative tensor also records only the input-symmetric part of the raw coefficient tensor. This is the differential version of Lemma [B.1](#).

C Expressivity proofs

C.1 Scalar-output reduction

Lemma C.1 (Output projection cannot increase channel rank). *For every $T \in \mathcal{T}_k$ and every $\eta \in (\mathbb{R}^p)^*$,*

$$r_{\text{tied}}((\eta \otimes I)T) \leq r_{\text{tied}}(T), \quad r_{\text{untied}}((\eta \otimes I)T) \leq r_{\text{untied}}(T).$$

Proof. Suppose first that T has a tied representation with m channels:

$$T = \sum_{a=1}^m w_a \otimes u_a^{\otimes k}.$$

Applying η to the output mode gives

$$(\eta \otimes I)T = \sum_{a=1}^m \eta(w_a) u_a^{\otimes k}.$$

This is a scalar tied representation with at most m atoms; atoms with $\eta(w_a) = 0$ simply drop out. Minimizing over tied representations of T gives the first inequality.

The detector-value case is identical. If

$$T = \sum_{a=1}^m w_a \otimes \text{Sym}(u_a^{\otimes(k-1)} \otimes v_a),$$

then

$$(\eta \otimes I)T = \sum_{a=1}^m \eta(w_a) \text{Sym}(u_a^{\otimes(k-1)} \otimes v_a),$$

which is a scalar detector-value representation with at most m atoms. $\square$

For the special embedding of a scalar target into one output coordinate, the ranks are exactly preserved.

Corollary C.1 (Scalar separations embed into vector targets). *Let $S \in \text{Sym}^k(\mathbb{R}^d)$, and let*

$$T = e_o \otimes S \in \mathbb{R}^p \otimes \text{Sym}^k(\mathbb{R}^d).$$

Then

$$r_{\text{tied}}(T) = r_{\text{tied}}(S), \quad r_{\text{untied}}(T) = r_{\text{untied}}(S).$$

Consequently, any scalar tied-detector-value separation transfers to the vector-valued CP-MLP setting.

Proof. The lower bounds follow from Lemma [C.1](#) by taking $\eta(y) = y_o$, since

$$(\eta \otimes I)T = S.$$

Thus

$$r_{\text{tied}}(S) \leq r_{\text{tied}}(T), \quad r_{\text{untied}}(S) \leq r_{\text{untied}}(T).$$

For the reverse inequalities, take any scalar representation of S and multiply each atom by the output basis vector e_o . This gives a vector-valued representation of $T = e_o \otimes S$ with the same number of channels. Hence the ranks are equal. $\square$

C.2 Bilinear atom-rank proofs

Proof of Theorem 4.5 We prove the tied and untied rank bounds separately.

Tied blocks. A tied bilinear block of width m represents a matrix of the form

$$M = \sum_{a=1}^m \alpha_a u_a u_a^\top.$$

Each term $\alpha_a u_a u_a^\top$ has rank at most one. Therefore

$$\text{rank}(M) \leq \sum_{a=1}^m \text{rank}(\alpha_a u_a u_a^\top) \leq m.$$

Since $M \in \mathbb{R}^{d \times d}$, also

$$\text{rank}(M) \leq d.$$

Thus a tied block satisfies

$$\text{rank}(M) \leq \min(m, d).$$

This bound is tight. Let $r = \min(m, d)$. Choosing

$$u_a = e_a, \quad \alpha_a = 1, \quad a = 1, \dots, r,$$

gives

$$M = \sum_{a=1}^r e_a e_a^\top,$$

which has rank $r = \min(m, d)$.

Untied blocks. An untied bilinear block of width m represents a matrix of the form

$$M = \sum_{a=1}^m M_a, \quad M_a = \frac{\alpha_a}{2} (u_a v_a^\top + v_a u_a^\top).$$

For any $z \in \mathbb{R}^d$,

$$M_a z = \frac{\alpha_a}{2} (u_a (v_a^\top z) + v_a (u_a^\top z)) \in \text{span}\{u_a, v_a\}.$$

Therefore

$$\text{rank}(M_a) \leq 2.$$

Using subadditivity of rank,

$$\text{rank}(M) \leq \sum_{a=1}^m \text{rank}(M_a) \leq 2m.$$

Since $M \in \mathbb{R}^{d \times d}$,

$$\text{rank}(M) \leq d.$$

Hence

$$\text{rank}(M) \leq \min(2m, d).$$

This bound is tight. Let $r = \min(2m, d)$.

If $r = 2s$ is even, choose

$$u_a = e_{2a-1}, \quad v_a = e_{2a}, \quad \alpha_a = 1, \quad a = 1, \dots, s.$$

Then

$$M = \frac{1}{2} \sum_{a=1}^s (e_{2a-1} e_{2a}^\top + e_{2a} e_{2a-1}^\top).$$

This is a direct sum of s independent 2×2 off-diagonal blocks, each of rank two. Therefore $\text{rank}(M) = 2s = r$.

If $r = 2s + 1$ is odd, use the same s off-diagonal blocks and add one diagonal atom by taking

$$u_{s+1} = v_{s+1} = e_{2s+1}, \quad \alpha_{s+1} = 1.$$

This adds one independent rank-one block, so the resulting matrix has rank $2s + 1 = r$. Thus the untied rank bound is tight. $\square$

Proof of Theorem 4.6 We prove the tied and untied atom ranks separately.

Tied atom rank. By Theorem 4.5, any tied representation of M with m atoms satisfies

$$\text{rank}(M) \leq m.$$

Therefore

$$r_{\text{tied}}(M) \geq \text{rank}(M).$$

For the reverse inequality, use the spectral decomposition of the symmetric matrix M :

$$M = \sum_{i=1}^{n_+(M)} \lambda_i q_i q_i^\top - \sum_{j=1}^{n_-(M)} \mu_j p_j p_j^\top, \quad \lambda_i, \mu_j > 0.$$

This is a tied representation using one signed rank-one atom for each nonzero eigendirection. Hence

$$r_{\text{tied}}(M) \leq n_+(M) + n_-(M) = \text{rank}(M).$$

Combining the two inequalities gives

$$r_{\text{tied}}(M) = \text{rank}(M) = n_+(M) + n_-(M).$$

Untied atom rank: lower bound. An untied atom satisfies the identity

$$\frac{\alpha}{2} (uv^\top + vu^\top) = \frac{\alpha}{4} (u+v)(u+v)^\top - \frac{\alpha}{4} (u-v)(u-v)^\top.$$

Therefore any width- m untied representation can be written as

$$M = AA^\top - BB^\top$$

for some $A, B \in \mathbb{R}^{d \times m}$, after absorbing scalar signs and magnitudes into the columns of A and B .

We claim that this implies

$$n_+(M) \leq m, \quad n_-(M) \leq m.$$

Indeed, on $\ker(A^\top)$,

$$x^\top Mx = x^\top (AA^\top - BB^\top)x = \|A^\top x\|^2 - \|B^\top x\|^2 = -\|B^\top x\|^2 \leq 0.$$

If $n_+(M) > \text{rank}(A)$, a subspace P of dimension $n_+(M)$ on which M is positive definite would satisfy $\dim(P \cap \ker A^\top) \geq n_+(M) - \text{rank}(A) > 0$, contradicting the inequality above; hence $n_+(M) \leq \text{rank}(A) \leq m$. Thus M cannot be positive definite on a subspace of dimension larger than $\text{rank}(A) \leq m$. Hence

$$n_+(M) \leq m.$$

Similarly, on $\ker(B^\top)$,

$$x^\top Mx = \|A^\top x\|^2 \geq 0,$$

so M cannot be negative definite on a subspace of dimension larger than $\text{rank}(B) \leq m$. Hence

$$n_-(M) \leq m.$$

Therefore every untied representation satisfies

$$m \geq \max\{n_+(M), n_-(M)\}.$$

So

$$r_{\text{untied}}(M) \geq \max\{n_+(M), n_-(M)\}.$$

Untied atom rank: upper bound. It remains to construct an untied representation with exactly $\max\{n_+(M), n_-(M)\}$ atoms. Use the spectral decomposition

$$M = \sum_{i=1}^{n_+(M)} \lambda_i q_i q_i^\top - \sum_{j=1}^{n_-(M)} \mu_j p_j p_j^\top, \quad \lambda_i, \mu_j > 0.$$

Pair positive and negative eigendirections as much as possible. For a paired block

$$\lambda q q^\top - \mu p p^\top, \quad \lambda, \mu > 0,$$

define

$$u = \sqrt{\lambda}q + \sqrt{\mu}p, \quad v = \sqrt{\lambda}q - \sqrt{\mu}p.$$

Then

$$\frac{1}{2}(uv^\top + vu^\top) = \lambda qq^\top - \mu pp^\top.$$

Thus one untied atom represents one positive and one negative eigendirection together.

If any positive eigendirections remain unpaired, represent each $\lambda qq^\top$ by the tied special case

$$u = v = \sqrt{\lambda}q.$$

If any negative eigendirections remain unpaired, represent each $-\mu pp^\top$ by

$$u = \sqrt{\mu}p, \quad v = -\sqrt{\mu}p.$$

Therefore the total number of atoms used is

$$\max\{n_+(M), n_-(M)\}.$$

This proves

$$r_{\text{untied}}(M) \leq \max\{n_+(M), n_-(M)\}.$$

Combining the lower and upper bounds gives

$$r_{\text{untied}}(M) = \max\{n_+(M), n_-(M)\}.$$

□

D Experimental details

This section documents the experimental methodology in sufficient detail for an independent re-implementation.

D.1 Computational requirements

All experiments run on a standard CPU; no GPU or specialized hardware is required. Each experiment completes in one to five minutes, and the full empirical program runs in approximately 15 minutes on a single CPU core. Memory usage stays below 1 GB throughout.

The experiments use only standard scientific Python: `numpy` for linear algebra, `scipy.optimize.minimize` (L-BFGS-B method) for nonlinear least-squares fitting, and `matplotlib` for figure generation. No deep-learning framework is required.

D.2 Data generation

The experiments fit two kinds of targets: *polynomial coefficient tensors* (in coefficient space) and *scalar functions of Gaussian inputs* (in function space). We describe both.

Coefficient-space targets. A degree- k polynomial in d variables corresponds to a coefficient vector $T \in \mathbb{R}^D$ in the basis of degree- k monomials, where $D = \binom{d+k-1}{k}$. For $d = 2$ we use the basis indexed by the exponent of x_2 , so index j corresponds to the monomial $x_1^{k-j}x_2^j$. The role-asymmetric witness $F_k = x_1^{k-1}x_2$ of Theorem 4.10 therefore has a single nonzero entry at index $j = 1$; the pure-power target x_1^k has its single nonzero entry at $j = 0$.

The model’s prediction in coefficient space is the coefficient vector of its parameterized polynomial, computed analytically from the architecture’s atom form (Section D.3). The loss is the squared ℓ_2 distance between the predicted coefficient vector and the target.

Function-space targets. For experiments using actual Gaussian inputs (Sections 5.3 and 5.4), we sample inputs $X \in \mathbb{R}^{N \times d}$ with each row drawn i.i.d. from $\mathcal{N}(0, I_d)$. The sample size N is given in Table 2. Targets are evaluated pointwise: for $y(x) = x_1^{k-1}x_2$, we set $y_i = X_{i,1}^{k-1}X_{i,2}$; for the indefinite quadratic experiment, $y_i = X_i^\top Q X_i$ with $Q = \text{diag}(\mathbf{1}^{n_+}, -\mathbf{1}^{n_-}, \mathbf{0}^{d-n_+-n_-})$. The loss is the mean squared error $\frac{1}{N} \sum_i (\hat{y}_i - y_i)^2$.

Random seeds. Each experiment uses random seeds that are deterministic functions of configuration values. For example, the quadratic experiment uses $\text{seed} = s * 137 + n_{\text{plus}} * 17 + n_{\text{minus}}$ for seed index s and signature (n_+, n_-) ; the width-sweep experiment uses $\text{seed} = s * 1000 + k * 100 + m$. This ensures full reproducibility across runs and makes results independent of seed-loop iteration order.

D.3 Model implementations

Each model is parameterized by a flat parameter vector $\theta \in \mathbb{R}^P$ that the optimizer treats as the only variable. The total parameter count P depends on the architecture and the width m .

Function-space architectures. Given input $X \in \mathbb{R}^{N \times d}$ and width m , all architectures follow the CP-MLP structure of Definition 3.1: project inputs through one or two linear maps, apply an elementwise nonlinearity, multiply per channel, and combine via scalar output coefficients $c \in \mathbb{R}^m$ (the scalar specialization of the output direction w_a).

For **tied** polynomial atoms with detector matrix $U \in \mathbb{R}^{m \times d}$ and coefficients c :

$$\hat{y}_i = \sum_{a=1}^m c_a (u_a^\top x_i)^k, \quad \theta = (\text{vec}(U), c) \in \mathbb{R}^{m(d+1)}.$$

For **untied** polynomial atoms with detector U , value $V \in \mathbb{R}^{m \times d}$, and coefficients c :

$$\hat{y}_i = \sum_{a=1}^m c_a (u_a^\top x_i)^{k-1} (v_a^\top x_i), \quad \theta = (\text{vec}(U), \text{vec}(V), c) \in \mathbb{R}^{m(2d+1)}.$$

For the four gated architectures in Section 5.4, replace the polynomial atoms with their gated analogues from Table I:

$$\begin{aligned} \text{tied ReLU}^2: \quad \hat{y} &= \sum_a c_a \text{ReLU}(u_a^\top x)^2, & \text{untied ReGLU}: \quad \hat{y} &= \sum_a c_a \text{ReLU}(u_a^\top x)(v_a^\top x), \\ \text{tied SiLU}^2: \quad \hat{y} &= \sum_a c_a \text{SiLU}(u_a^\top x)^2, & \text{untied SwiGLU}: \quad \hat{y} &= \sum_a c_a \text{SiLU}(u_a^\top x)(v_a^\top x), \end{aligned}$$

with $\text{SiLU}(z) = z\sigma(z)$ and $\sigma(z) = 1/(1 + e^{-z})$. Tied gated architectures have parameter count $m(d+1)$; untied gated architectures have $m(2d+1)$. The matched-parameter control of Section 5.4 pairs untied width m against tied width $2m$, equalizing input-projection parameters at $2md$.

Coefficient-space implementations. For coefficient-space experiments we compute the coefficient vector of each parameterized polynomial directly, avoiding sampling. For tied atoms, the multinomial theorem gives

$$(u^\top x)^k = \sum_{|\alpha|=k} \binom{k}{\alpha} u^\alpha x^\alpha, \quad \binom{k}{\alpha} = \frac{k!}{\alpha_1! \cdots \alpha_d!}.$$

For untied detector-value atoms,

$$(u^\top x)^{k-1} (v^\top x) = \left(\sum_{|\beta|=k-1} \binom{k-1}{\beta} u^\beta x^\beta \right) \sum_{j=1}^d v_j x_j,$$

which expands into a degree- k multi-index sum. The total predicted coefficient vector is $\sum_a c_a$ times the per-atom coefficient vector, and the loss is the squared ℓ_2 distance from the target.

D.4 Optimization protocol

All experiments minimize the squared loss using `scipy.optimize.minimize` with method L-BFGS-B, tolerances `ftol` = 10^{-22} and `gtol` = 10^{-15} , and `maxiter` $\in [2000, 3000]$ depending on experiment. For each random seed, we run multiple restarts with parameter initialization $\theta_0 \sim \mathcal{N}(0, 0.25)$ (applied independently to each entry of θ). We record the best-of-restarts loss per seed and aggregate (median, min, max) across seeds.

Table 2: Hyperparameters for each experiment.

Experiment	Section	Input dim d	Sample size	Seeds	Restarts
Indefinite quadratic targets	§5.1	8	256	3	8
Role-asymmetric monomial targets	§5.2	2	coef. space	3	10
Pure-vs-mixed reversal	§5.3	4	256	10	5
Gated MLPs	§5.4	4	200	3	3
Matched-parameter control (Gated MLPs)	§5.4	4	200	3	3

Table 3: Scope boundary at $k = 3$. Median best-of-restarts MSE over three seeds.

Target	Architecture	$m = 1$	$m = 2$	$m = 3$	$m = 4$
$x_1^{k-1}x_2$ (role-asym.)	tied	4.0×10^{-1}	2.0×10^{-8}	2.3×10^{-10}	1.2×10^{-13}
	untied (det.-val.)	$< 10^{-15}$	6.3×10^{-15}	2.0×10^{-15}	$< 10^{-15}$
	multi-leg oracle	1.2×10^{-13}	6.4×10^{-14}	$< 10^{-15}$	$< 10^{-15}$
$x_1x_2 \cdots x_k$ (square-free)	tied	6.1×10^{-1}	3.3×10^{-1}	1.4×10^{-1}	1.2×10^{-15}
	untied (det.-val.)	3.3×10^{-1}	$< 10^{-15}$	$< 10^{-15}$	$< 10^{-15}$
	multi-leg oracle	$< 10^{-15}$	$< 10^{-15}$	$< 10^{-15}$	$< 10^{-15}$

D.5 Per-experiment configurations

For coefficient-space experiments, fits operate directly on the $\binom{d+k-1}{k}$ monomial coefficients of the target polynomial, eliminating sampling and function-evaluation noise. For experiments with finite samples, inputs are drawn from $\mathcal{N}(0, I_d)$ once per seed and held fixed across restarts within that seed.

D.6 Convergence criterion

We report best-of-restarts MSE without applying a separate convergence threshold. A configuration is described as *fitting* the target if the residual reaches machine precision ($\text{MSE} < 10^{-8}$) and as *failing* if the residual stays above 10^{-1} . The intermediate regime is interpreted as a partial fit due to optimizer-landscape difficulty rather than a representational limitation, and is discussed for the $k = 5$ case in Section 5.2.

To distinguish the plateau of Section 5.2 from a slow-optimization artifact, we re-ran the failed configurations with $5 \times$ the L-BFGS iteration budget. The residual stayed within 10% of its plateau value across all configurations tested, confirming that the plateau is robust to optimization budget.

D.7 Scope of the role-asymmetric separation

Theorem 4.10 shows that one untied detector-value atom suffices for the role-asymmetric witness $F_{dv} = x_1^{k-1}x_2$, where one direction acts as a repeated detector and one direction supplies the single value. This appendix tests the boundary of that statement: targets with multiple independent value roles do *not* reduce to a single untied atom.

Concretely, at $k = 3$ we compare tied, untied, and a multi-leg oracle (the smallest atom family rich enough to express any product of k linear forms) on the role-asymmetric witness F_{dv} and on the square-free product

$$F_{sf} = x_1x_2 \cdots x_k,$$

in which each of the k variables appears exactly once.

Figure 6 and Table 3 confirm three regimes. On F_{dv} , untied atoms and the multi-leg oracle both fit at $m = 1$, while tied requires $m \geq k$. On F_{sf} , the multi-leg oracle fits at $m = 1$, untied requires $m = 2$, and tied requires $m \geq 2^{k-1}$, matching its Waring rank (by Proposition 4.9). The untied advantage is therefore specialized to targets with a single value role and does not extend to arbitrary products.

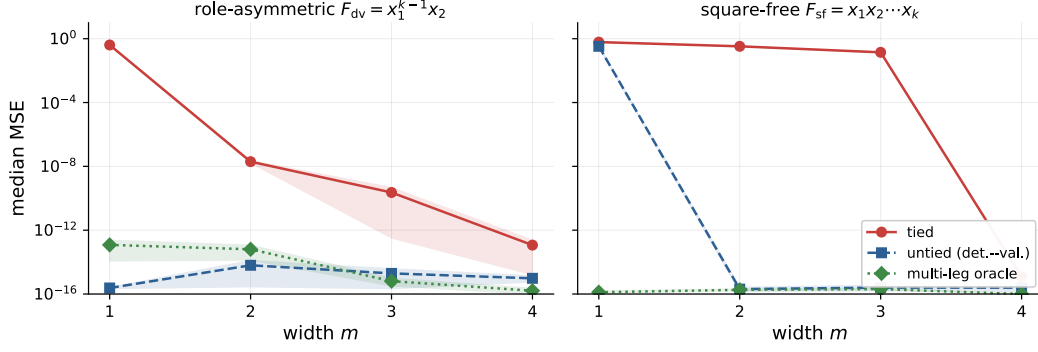


Figure 6: Scope boundary at $k = 3$. Untied atoms are rank one on the role-asymmetric witness $F_{dv} = x_1^{k-1}x_2$ but not on the square-free product $F_{sf} = x_1x_2 \cdots x_k$; the multi-leg oracle is rank one on both.

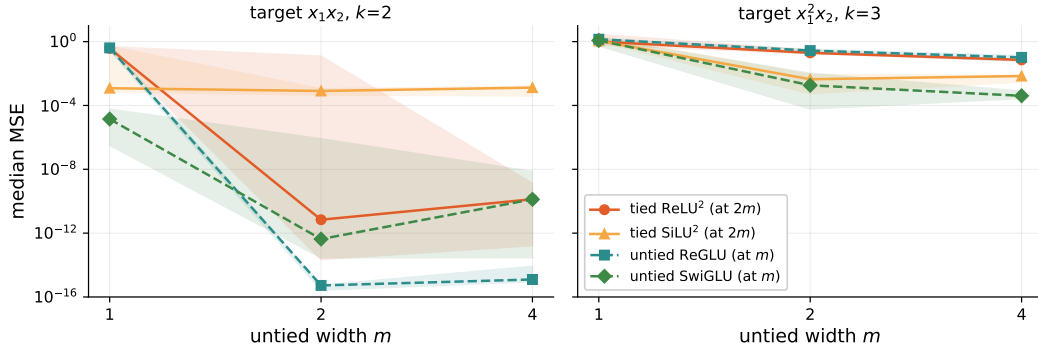


Figure 7: Matched-parameter control on the role-asymmetric target $x_1^{k-1}x_2$. Untied architectures at width m (dashed) still beat tied architectures at width $2m$ (solid), so input-projection parameter counts match. Median over three seeds; bands show min-max.

D.8 Matched-parameter control in gated MLPs

A natural concern about Section 5.4 is that untied architectures use two input projections (U, V) per channel and therefore carry more parameters than tied architectures at the same width. To rule out parameter count as the driver of the gap, we pair untied at width m against tied at width $2m$, which equalizes input-projection parameters at $2md$ entries.

Figure 7 shows that the untied advantage *persists* under this control. At $k = 2$, untied ReGLU at $m = 2$ reaches 5.7×10^{-16} , while tied ReLU^2 at $m = 4$ reaches only 4.9×10^{-11} —roughly five orders of magnitude. At $k = 3$, untied SwiGLU at $m = 4$ beats tied SiLU^2 at $m = 8$ by approximately one order of magnitude. The detector-value advantage is therefore not absorbed by giving tied blocks twice as many channels.